

# Expected Cost and Degenerate Cases

## 1 Know Where You Are

- the planar point location problem

## 2 Expected Cost

- statement of the theorem
- outline of the ideas in the proof
- expected depth of the directed acyclic graph

## 3 Degenerate Cases

- vertical line segments
- query points on extensions or segments

MCS 481 Lecture 20  
Computational Geometry  
Jan Verschelde, 9 October 2026

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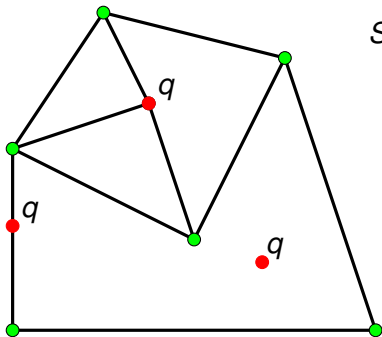
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# the planar point location problem

Given a map, do you know where you are?

Input: a planar subdivision  $S$ ,  $n = \# \text{edges of } S$ ,  
a query point  $q = (q_x, q_y)$ .

Output: face of  $S$  that contains  $q$ , or eventually  
the edge which contains  $q$ , or the vertex matching  $q$ .

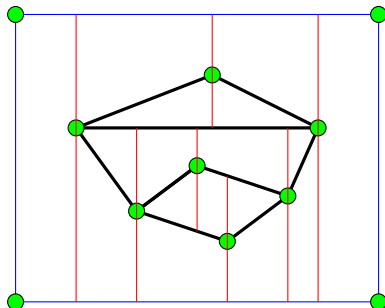
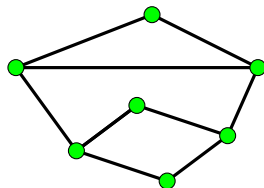


$S, n = 10$

3 cases:

- $q$  in face
- $q$  on edge
- $q$  is vertex

# a trapezoidal decomposition of a subdivision



Constructing a trapezoidal decomposition  $T(S)$  of a subdivision  $S$ :

for every edge  $e$  in  $S$  do

for every end point  $p$  of  $e$  do

add a vertical extension from  $p$  to the next edge below  $p$

add a vertical extension from  $p$  to the next edge above  $p$

# faces and storage cost

In lecture 18, we proved two lemmas.

## Lemma (trapezoidal faces)

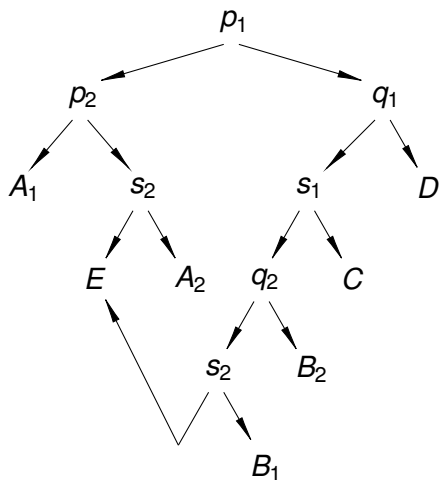
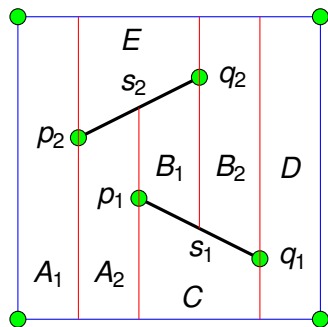
*Every face in a trapezoidal decomposition  $T(S)$  has one or two vertical sides and exactly two skew sides.*

## Lemma (linear storage cost)

*Let  $S$  be a subdivision with  $n$  edges and bounding box  $R$ . Its trapezoidal decomposition  $T(S)$  has at most  $6n + 4$  vertices and at most  $3n + 1$  trapezoids.*

We might still hope for a linear time algorithm ...

# a trapezoidal map and its search structure



In the directed acyclic graph, every trapezoid is stored only once.

# a randomized incremental algorithm

## Algorithm TRAPEZOIDALMAP( $S$ )

Input:  $S$  is a subdivision with  $n$  line segments.

Output:  $T(S)$  is a list of trapezoids,

$\mathcal{D}$  a directed acyclic graph to search  $T(S)$ .

- 1  $R = \text{BOUNDINGBOX}(S)$
- 2  $T(S_0) = R; \mathcal{D} = \text{VERTICES}(R)$
- 3  $s_1, s_2, \dots, s_n = \text{PERMUTE}(S)$
- 4 for  $i = 1, 2, \dots, n$  do
- 5      $\Delta_0, \Delta_1, \dots, \Delta_k = \text{FOLLOWSEGMENT}(T(S_{i-1}), \mathcal{D}, s_i)$
- 6      $\Delta'_0, \Delta'_1, \dots, \Delta'_\ell = \text{UPDATET}(T(S_{i-1}), \Delta_0, \Delta_1, \dots, \Delta_k, s_i)$
- 7      $\text{UPDATED}(\mathcal{D}, \Delta_0, \Delta_1, \dots, \Delta_k, \Delta'_0, \Delta'_1, \dots, \Delta'_\ell, s_i)$

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# statement of expected cost

Given is a subdivision  $S$  with  $n$  edges.

In the randomized algorithm, there are  $n!$  ways to add the edges.

The expected cost is the average cost over all  $n!$  running times.

## Theorem (expected cost of a randomized incremental algorithm)

*Given is a subdivision  $S$  with  $n$  edges in general position.*

- 1 *A trapezoidal map  $T(S)$  and a search structure  $\mathcal{D}$  can be computed in  $O(n \log(n))$  expected time.*
- 2 *The expected size of the search structure  $\mathcal{D}$  is  $O(n)$ .*
- 3 *For any query point  $q$ , the expected query time is  $O(\log(n))$ .*

The proof is based on a probabilistic analysis.

## two lemmas imply the result

### Lemma (expected query time)

*The expected query time is  $O(\log(n))$ .*

### Lemma (expected size of search structure)

*The expected size of the search structure is  $O(n)$ .*

Then the overall running time of the algorithm is

$$O(n) + \sum_{i=1}^n \left[ \underbrace{O(\log(i))}_{\text{search}} + \underbrace{\frac{O(i)}{i}}_{\text{size}} \right] = O(n \log(n)).$$

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# expected query time

## Lemma (expected query time)

*The expected query time is  $O(\log(n))$ .*

- 0 Worst case:  $3n$  is an upper bound on path length.
- 1  $X_i$  is the number of nodes added to path by  $s_i$ , and

$$P_i = \text{Prob}[\Delta_q(s_i) \neq \Delta_q(s_{i-1})], \quad E[X_i] \leq 3P_i.$$

The expected value of  $X_i$  depends on the probability that  $s_i$  intersects  $\Delta$ .

- 2  $E \left[ \sum_{i=1}^n X_i \right] = 12H_n$  is  $O(\log(n))$ , where

$$H_n = \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n} \text{ is the } n\text{-th Harmonic number.}$$

## expected size

### Lemma (expected size of search structure)

*The expected size of the search structure is  $O(n)$ .*

0 Worst case:  $O(n) + \sum_{i=1}^n O(i) = O(n^2)$ , #leaves  $\leq 3n + 1$ .

- 1 Let  $k_i$  be the number of new leaves in the  $i$ -th iteration.  
Then the number of inner nodes is  $k_i - 1$ .  
For  $S_i \subseteq S$ ,  $\Delta \in T(S_i)$ ,  $s \in S_i$ , consider:

$$\delta(\Delta, s) = \begin{cases} 1 & \text{if } s \notin S_i \Rightarrow \Delta \notin T(S_i), \\ 0 & \text{otherwise.} \end{cases}$$

- 2 Take the average over all segments  $s_1, s_2, \dots, s_i$ :

$$E[k_i] = \frac{1}{i} \sum_{s \in S_i} \sum_{\Delta \in T(S_i)} \delta(\Delta, s) \leq \frac{4}{i} |T(S_i)| = \frac{O(i)}{i}.$$

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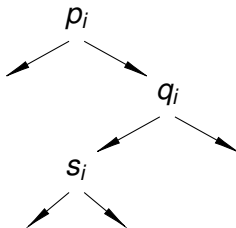
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## expected depth of the directed acyclic graph

The depth of the directed acyclic graph  $\mathcal{D}$  determines the search cost. In the worst case, every segment  $s_i(p_i, q_i)$  adds 3 nodes:



For  $n$  segments,  $3n$  is an upper bound on the path length in  $\mathcal{D}$ .

The random variable  $X_i$  equals the number of nodes added to the path length of  $\mathcal{D}$  when inserting  $s_i$ .

Linearity of expected values: 
$$E \left[ \sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i].$$

# the probability of a new node

## Definition (probability of new node at iteration $i$ )

Let  $q$  be a query point. The **probability**  $P_i$  is the probability there exists a node on the search path of  $q$  created at iteration  $i$ .

Because at most 3 nodes are created in iteration  $i$ :  $E[X_i] \leq 3P_i$ .

Let  $\Delta_q(S_i)$  be the  $\Delta \in T(S_i)$ :  $q \in \Delta$ .

If  $\Delta_q(S_i) \neq \Delta_q(S_{i-1})$ , then  $\Delta_q(S_i)$  is created in iteration  $i$ .

Therefore,  $P_i = \text{Prob}[\Delta_q(S_i) \neq \Delta_q(S_{i-1})]$ .

All trapezoids  $\Delta$  created in iteration  $i$  are adjacent to  $s_i$ :

- 1 either  $\text{top}(\Delta)$  or  $\text{bottom}(\Delta)$  is  $s_i$ ,
- 2 or  $\text{leftp}(\Delta)$  or  $\text{rightp}(\Delta)$  is an endpoint of  $s_i$ .

# backwards analysis

All trapezoids  $\Delta$  created in iteration  $i$  are adjacent to  $s_i$ :

- 1 either  $\text{top}(\Delta)$  or  $\text{bottom}(\Delta)$  is  $s_i$ ,
- 2 or  $\text{leftp}(\Delta)$  or  $\text{rightp}(\Delta)$  is an endpoint of  $s_i$ .

Then, observe:

- 1 the probability that  $s_i$  happens to be  $\text{top}(\Delta)$  is  $1/i$ ,
- 2 the probability that  $s_i$  happens to be  $\text{bottom}(\Delta)$  is  $1/i$ ,
- 3 the probability that an end point of  $s_i$  is  $\text{leftp}(\Delta)$  equals  $1/i$ ,
- 4 the probability that an end point of  $s_i$  is  $\text{rightp}(\Delta)$  equals  $1/i$ .

$$\begin{aligned}\Rightarrow P_i &= \text{Prob}[\Delta_q(S_i) \neq \Delta_q(S_{i-1})] \\ &= \text{Prob}[\Delta_q(S_i) \notin T(S_{i-1})] \\ &\leq 4/i.\end{aligned}$$

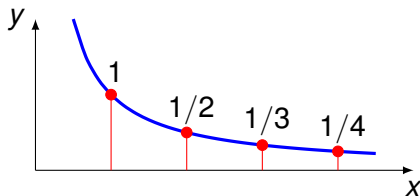
# Harmonic numbers

$X_i$  is the number of nodes added to the path length of  $\mathcal{D}$  at iteration  $i$ .

$$E \left[ \sum_{i=1}^n X_i \right] \leq \sum_{i=1}^n 3P_i \leq \sum_{i=1}^n \frac{12}{i} = 12 \sum_{i=1}^n \frac{1}{i} = 12H_n$$

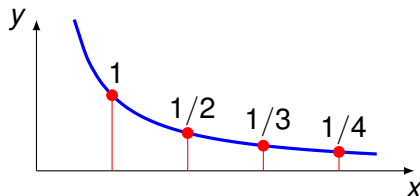
$H_n = \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n}$  is the  $n$ -th Harmonic number.

Consider the plot of  $y = 1/x$ .



# the average path length

Consider the plot of  $y = 1/x$ .



$H_n = \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n}$  relates to the area under  $y = 1/x$ :

$$\int \frac{1}{x} dx = \ln(x) + C, \quad \text{for some constant } C.$$

Apply the forward integration rule to obtain an upper bound:  
 $H_n \leq \ln(n) + 1$ . Therefore: the average path length is  $O(\log(n))$ .

# Riemann sums

## Exercise 1:

Consider the definite integral

$$\int_1^{n+1} \frac{1}{x} dx = \ln(n+1).$$

- 1 Apply the Riemann sum to the integral and relate the value of the sum to  $H_n$ .
- 2 As  $n \rightarrow \infty$ ,  $H_n$  diverges (slowly). Consider

$$\lim_{n \rightarrow \infty} (H_n - \ln(n)) = \gamma,$$

where  $\gamma$  is a constant.

Run numerical experiments to approximate  $\gamma$ .

For  $k$  ranging from 1 to 10, how large should  $n$  be to obtain  $k$  correct decimal places of  $\gamma$ ?

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# assumptions on the input

We assumed that segments are *in general position*:

- 1 All points lie in a bounding box.
- 2 There are no vertical segments.

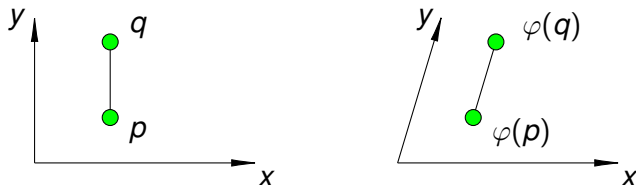
The absence of vertical segments limited the number of adjacent trapezoids to four.

Furthermore, what if query points

- 1 lie on a vertical extension?
- 2 lie on a line segment?

# a shear transformation

Consider tilting the vertical axis:



The new coordinates are computed with a *shear transformation*  $\varphi$ :

$$\begin{aligned}\varphi : \quad \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ (x, y) &\mapsto (x + \epsilon y, y), \quad \epsilon > 0.\end{aligned}$$

# the effectiveness of the shear transformation

$$\begin{aligned}\varphi : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ (x, y) &\mapsto (x + \epsilon y, y), \quad \epsilon > 0.\end{aligned}$$

Consider  $p$  and  $q$ , as endpoints of a vertical segment:



Let  $p = (a, b_1)$  and  $q = (a, b_2)$ ,  $b_1 < b_2$ :

$$\varphi(p) = (a + \epsilon b_1, b_1) \quad \text{and} \quad \varphi(q) = (a + \epsilon b_2, b_2).$$

$$b_1 < b_2 \quad \Rightarrow \quad a + \epsilon b_1 < a + \epsilon b_2,$$

so  $\varphi(q)$  is at the right of  $\varphi(p)$ . Comparisons are done *symbolically*.

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# query points on extensions

In the directed acyclic graph we have two types of nodes:

- 1 at  $x$ -node, decide whether at left or right of endpoint,
- 2 at  $y$ -node, decide whether above or below a segment.

What if query point  $q$  lies on vertical extension or line segment?

- 1 At  $x$ -node, we compare lexicographically.

Equality will hold only if  $q$  coincides with an endpoint.

**Exercise 2:** Verify the above statement.

# query points on segments

In the directed acyclic graph we have two types of nodes:

- 1 at  $x$ -node, decide whether at left or right of endpoint,
- 2 at  $y$ -node, decide whether above or below a segment.

What if query point  $q$  lies on vertical extension or line segment?

- 2 At  $y$ -node, when does  $q$  lie on the segment  $s$ ?

If the transformed point  $\varphi(q)$  lies on  $\varphi(s)$ ,  
then also the original point  $q$  lies on  $s$ .

**Exercise 3:** Verify the above statement.

# the main result

With the modifications to the Algorithm TRAPEZOIDALMAP so it also works for special, degenerate cases, we have:

## Theorem (expected cost of a randomized incremental algorithm)

*Given is a subdivision  $S$  with  $n$  edges ~~in general position~~.*

- 1 *A trapezoidal map  $T(S)$  and a search structure  $\mathcal{D}$  can be computed in  $O(n \log(n))$  expected time.*
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# summary and exercises

We finished chapter 6 in the textbook.

Consider the following activities, listed below.

- 1 Write the solutions to exercises 1, 2, and 3.
- 2 Consult the CGAL documentation and example code on trapezoidal maps.
- 3 Consider the exercises 11, 13, 15 in the textbook.